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Solve The Following system of
First-order linear difference
equations:

$$x_t = -4x_{t-1} + y_{t-1} + 12 \quad ; \quad x_0 = 16$$

$$y_t = 2x_{t-1} - 3y_{t-1} + 6 \quad y_0 = 8$$

Set them in matrix form,

$$\begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} 12 \\ 6 \end{bmatrix}$$

$$X_t = AX_{t-1} + B$$

we next find the complementary functions
assuming a case distinct real
roots: $y_c = k_1 C_1 r_1^t + k_2 C_2 r_2^t$
on the characteristics roots are

$$r_1, r_2 = \frac{\text{Tr}(A) \pm \sqrt{[\text{Tr}(A)]^2 - 4|A|}}{2}$$

$$\text{Tr}(A) = -4 - 3 \Rightarrow -7$$

$$|A| = 12 - 2 \Rightarrow 10$$

$$\text{Substituting: } r_1, r_2 = \frac{-7 \pm \sqrt{(-7)^2 - 4(10)}}{2}$$

$$r_1, r_2 = \frac{-7 \pm 3}{2} \Rightarrow r_1 = \frac{-7+3}{2} = -2$$

$$r_2 = \frac{-7-3}{2} = -5$$

$$\lambda_1 = -2, \quad \lambda_2 = -5$$

characteristics roots as eigenvalues:

3 we then find the eigenvalues

using secularity $(A - \lambda_i I)$ is singular

$$(A - \lambda_i I)C_i = \begin{bmatrix} a_{11} - \lambda_i & a_{12} \\ a_{21} - \lambda_i & a_{22} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

a subtracting first from for $\lambda_1 = -2$,

$$\begin{bmatrix} -4 - (-2) & 1 \\ 2 & -3 - (-2) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

Then by multiplying row by column

we find: $-2c_1 + c_2 = 0$; $c_2 = 2c_1$

$$2c_1 + c_1 = 0; \quad c_2 = 2c_1$$

$$c_1 = 1 \quad c_2 = 2$$

Hence the eigenvector corresponding

to: $\lambda_1 = -2$ is

$$C_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$K_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-2t} = \begin{bmatrix} K_1 (-2)^t \\ 2K_1 (-2)^t \end{bmatrix}$$

b Then subtracting from $\lambda_2 = -5$

$$\begin{bmatrix} -4 - (-5) & 1 \\ 2 & -3 - (-5) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

Multiplying row by column,

$$c_1 + c_2 = 0, \quad c_2 = -c_1$$

$$2c_1 + 2c_2 = 0$$

at $c_1 = 1$

, $c_2 = -1$

Then eigenvector for $\lambda_2 = -5$ is

$$c_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

and the

$$k_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} (-5)^t = \begin{bmatrix} k_2 (-5)^t \\ -k_2 (-5)^t \end{bmatrix}$$

$$x_c = k_1 (-2)^t + k_2 (-5)^t$$

$$y_c = 2k_1 (-2)^t + k_2 (-5)^t$$

Now we find the particular solution for the steady-state solution $\bar{x}, \bar{y}$; $\bar{y} = (\bar{I} - A)^{-1} B$ where

$$(\bar{I} - A) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -4 & 1 \\ 2 & -3 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 & 1 \\ -2 & 4 \end{bmatrix}$$

$$y_p = \begin{bmatrix} \bar{x} \\ \bar{y} \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 4 & 1 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 12 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

This make the complete general solution.

$$x_t = k_1 (-2)^t + k_2 (-5)^t + 3$$

$$y_t = 2k_1 (-2)^t + k_2 (-5)^t + 3$$

with $-2 > 1$ and $-5 > 1$, the time path is divergent.

Example 6

For the reduce to

$$k_1 + k_2 + 3 = 16$$

$$2k_1 - k_2 + 3 = 8$$

Simultaneously, $k_1 = 6$, $k_2 = 7$.

by simple substitution, we then find specific solution:

$$x_t = 6(-2)^t + 7(-5)^t + 3$$

$$y_t = 12(-2)^t - 7(-5)^t + 3$$

To check answers substitute, ~~we~~ then find the specific solution: $t = 1$ and $t = 0$

$$x_1 = 6(-2)^1 + 7(-5)^1 + 3 = -44$$

$$y_1 = 12(-2)^1 - 7(-5)^1 + 3 = 14$$

$$x_0 = 6(-2)^0 + 7(-5)^0 + 3 = 16$$

$$y_0 = 12(-2)^0 - 7(-5)^0 + 3 = 8$$

x_1, y_1 for x_2 and y_2 and x_0, y_0 for x_{t-1} and y_{t-1}

$$-44 = -4(16) + 8(12) \Rightarrow -44$$

$$14 = 2(16) - 3(8) + 6 \Rightarrow 14$$

Example: 7

Solve the

equations:

$$x_t = 4x_{t-1} - 2y_{t-1} + y_t = 10 \quad ; \quad x_0 = 20$$

$$y_t = 3x_{t-1} + 6y_{t-1} - 4 \quad ; \quad y_0 = 3$$

Rearrange the system and set in

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ y_{t-1} \end{bmatrix} + \begin{bmatrix} 10 \\ -4 \end{bmatrix}$$

$$A_1 Y_t = A_2 Y_{t-1} + B$$

For the complementary function:

$$(A_2 - \lambda I A_1) = 0 \quad \text{and substitute}$$

The parameters of the problems,

$$|A_2 - \lambda I A_1| = \begin{vmatrix} 4-\lambda & -2+\lambda \\ 3 & 6-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)(6-\lambda) - 3(-2+\lambda) = 0$$

$$\lambda^2 - 13\lambda + 30 = 0 \quad ; \quad \lambda_1 = 3, \quad \lambda_2 = 10$$

Find the nontrivial solutions for

the eigenvalues λ_i

$$(A_2 - \lambda_i I A_1) c_i = 0$$

where:

$$(A_2 - \lambda_i A_1) c_i = \begin{bmatrix} 4-\lambda_i & 2+\lambda_i \\ 3 & 6-\lambda_i \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Subtracting for $\lambda_1 = 3$

$$\begin{bmatrix} 4-3 & -2+3 \\ 3 & 6-3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

Multiplying row by column:

" $c_2 = 1$ "

$$c_1 + c_2 = 0, \quad 3c_1 + 3c_2 = 0$$

$$c_1 + 1 = 0, \quad 3c_1 + 3(1) = 0$$

$$c_1 = -1, \quad 3c_1 + 3 = 0$$

$$3c_1 = -3$$

$$c_1 = \frac{-3}{3} = -1$$

$$-1 + c_2 = 0$$

$$c_2 = 1$$

Q1 $c_1 = 1, c_2 = -1$, the eigenvector

$$c_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Making the elements of a complementary function for $\lambda_1 = 3$,

$$k_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} (3)^t = \begin{bmatrix} k_1 (3)^t \\ -k_1 (3)^t \end{bmatrix}$$

b For $\lambda_2 = 10$

$$\begin{bmatrix} 4-10 & -2+10 \\ 3 & 6-10 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -6 & 8 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = 0$$

By matrix multiplication,

$$-6c_1 + 8c_2 = 0 \quad 3c_2 - 4c_1 = 0$$

Letting $c_1 = 1$, $c_2 = 0.75$ the corresponding eigenvector becomes

$$c_1 = \begin{bmatrix} 1 \\ 0.75 \end{bmatrix}$$

and the complementary

k_2

complementary

Now

steady

$y_p =$

where

The

$y_0 =$

Since

system

is not

and the all elements of the complementary function for $x_2 = 10$

$$x_2 \begin{bmatrix} 1 \\ 0.75 \end{bmatrix} (10)^t = \begin{bmatrix} x_2 (10)^t \\ 0.75 x_2 (10)^t \end{bmatrix}$$

Combining two function, the complementary function becomes

$$x_c = k_1 (3)^t + k_2 (10)^t$$

$$x_c = -k_1 (3)^t + 0.75 k_2 (10)^t$$

Now find the particular or steady ~~state~~ state solution
 $x_p = \bar{y}$

$$\bar{y} = (A_1 - A_2)^{-1} B$$

$$\text{where } A_1 - A_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 4 & -2 \\ 3 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} -3 & 1 \\ -3 & -5 \end{bmatrix}$$

The complete general solution

$x_0 = x_c + x_p$, then becomes

$$x_t = k_1 (3)^t + k_2 (10)^t + 3$$

$$x_t = -k_1 (3)^t + 0.75 k_2 (10)^t - 1$$

Since $x_1 = 3$, $x_2 = 10 > 11$, the system of equation is dynamically unstable.

Example: 8

The specific solution at $t=0$,

$$k_1 + k_2 + 3 = 20$$

$$-k_1 + 0 - 75k_2 - 1 = 3$$

Solving simultaneously:

$$k_1 = 5, \quad k_2 = 12$$

and by substituting, we come to the definite solution

$$x_t = 5(3)^t + 12(10)^t + 3$$

$$x_t = -5(3)^t + 9(10)^t - 1$$

To check the answers substitute:

$$t=1 \quad \text{and} \quad t=0$$

$$x_1 = 5(3) + 12(10) + 3 = 138$$

$$x_1 = -5(3) + 9(10) - 1 = 74$$

$$x_0 = 5 + 12 + 3 = 20$$

$$y_0 = -5 + 9 - 1 = 3$$

Substitute x_1, y_1 for x , and y_1 for

x_0, y_0 for x_{t-1}, y_{t-1}

$$138 = 4(20) - 2(3) + 74 - 10 = 138$$

$$74 = 3(20) + 6(3) - 4$$

$$74 = 74$$